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(a)	The internal energy of a body is the sum of the random distribution of all the potential and kinetic energies of the microscopic particles / atoms / molecules in the gas.
(b)(i)	Since B and C are at the same temperature, $P_C V_C = P_B V_B$
	$P_B = 6.36 \times 10^4 \text{ Pa}$
	Alternative Using Ideal gas law, $P_B V_B = nRT_B$
	$P_B = 6.35 \times 10^4 \text{ Pa}$

(b)(ii)	Applying Ideal gas equation, $\Delta U_{CD} = \frac{3}{2} nR\Delta T = \frac{3}{2} \Delta(PV) = \frac{3}{2} V(\Delta P)$																				
	$\Delta U_{CD} = -11.2 \times 10^{-3} \text{ J}$																				
(b)(iii)	Isovolumetric process: $W = 0 \text{ J}$ $W_{CD} = W_{AB} = 0 \text{ J}$ For a cyclic process, $\Delta U = 0$ $\Rightarrow \Delta U_{AB} = -\Delta U_{CD} = 11.2 \times 10^{-3} \text{ J}$ Apply first law of thermodynamics $Q_{AB} = 11.2 \times 10^{-3} \text{ J}; Q_{BC} = -27.5 \times 10^{-3} \text{ J};$ $Q_{CD} = -11.2 \times 10^{-3} \text{ J}; Q_{DA} = 21.6 \times 10^{-3} \text{ J}$ <table><tr><td></td><td>heat supplied / $\times 10^{-3} \text{ J}$</td><td>work done on the gas / $\times 10^{-3} \text{ J}$</td><td>change in internal energy / $\times 10^{-3} \text{ J}$</td></tr><tr><td>A $\rightarrow$ B</td><td>11.2</td><td>0</td><td>11.2</td></tr><tr><td>B $\rightarrow$ C</td><td>-27.5</td><td>27.5</td><td>0</td></tr><tr><td>C $\rightarrow$ D</td><td>-11.2</td><td>0</td><td>-11.2</td></tr><tr><td>D $\rightarrow$ A</td><td>21.6</td><td>-21.6</td><td>0</td></tr></table>		heat supplied / $\times 10^{-3} \text{ J}$	work done on the gas / $\times 10^{-3} \text{ J}$	change in internal energy / $\times 10^{-3} \text{ J}$	A $\rightarrow$ B	11.2	0	11.2	B $\rightarrow$ C	-27.5	27.5	0	C $\rightarrow$ D	-11.2	0	-11.2	D $\rightarrow$ A	21.6	-21.6	0
	heat supplied / $\times 10^{-3} \text{ J}$	work done on the gas / $\times 10^{-3} \text{ J}$	change in internal energy / $\times 10^{-3} \text{ J}$																		
A $\rightarrow$ B	11.2	0	11.2																		
B $\rightarrow$ C	-27.5	27.5	0																		
C $\rightarrow$ D	-11.2	0	-11.2																		
D $\rightarrow$ A	21.6	-21.6	0																		
(b)(iv)1.	D $\rightarrow$ A																				
(b)(iv)2.	$W_{net} = 27.5 \times 10^{-3} + (-21.6 \times 10^{-3})$																				
	$\text{COP} = \frac{21.6 \times 10^{-3}}{5.9 \times 10^{-3}}$ $= 3.7 \text{ (2sf)}$																				
4																					